Exp No: 02

Determination and plotting of Electric field and Electric flux density in sphere full of charges with varying radius.

Aim:

To write a program for electric field and electric flux density in sphere using SCILAB.

Software required:

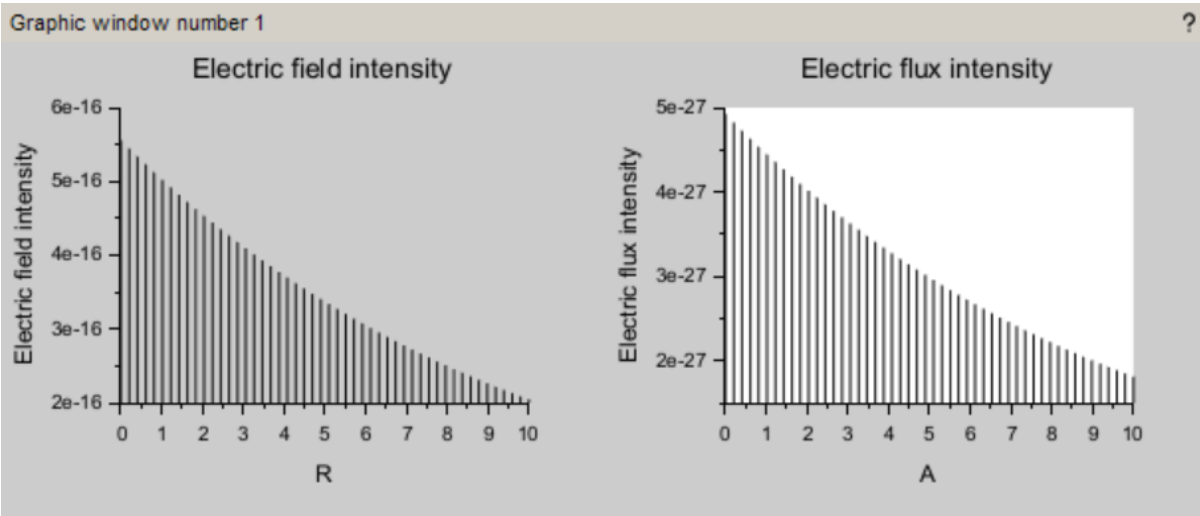
- SCILAB version 6.0.1

Program

```
clc;
clear;
q=input('Enter the value of q=');
r=input('Enter the value of r=');
efcilon=8.854*10^-12;
pi=3.14
E=q/4*pi*efcilon*r^2;
disp("The electric field is", [E]);
D=E*efcilon;
disp("The flux density is", [D]);
x = linspace(0, 10, 50);
y = E * exp(-0.1 * x);
z = D * exp(-0.1 * x);
figure();
subplot(2, 2, 1);
plot2d3(x,y);
xlabel("R");
ylabel("Electric field intensity");
title('Electric field intensity');
subplot(2, 2, 2);
plot2d3(x,z);
xlabel("A");
ylabel("Electric flux intensity");
title('Electric flux intensity');
```

Output :

```
Enter the value of q=5*10^-6

Enter the value of r=4

  "The electric field is"
   5.560D-16
  "The flux density is"
   4.923D-27
```



Case 1:

Derive and plot the electric field (E) and electric flux density (D) inside a uniformly charged sphere with charge density ρ0. What are the values of E and D at the center and at the surface of the sphere?

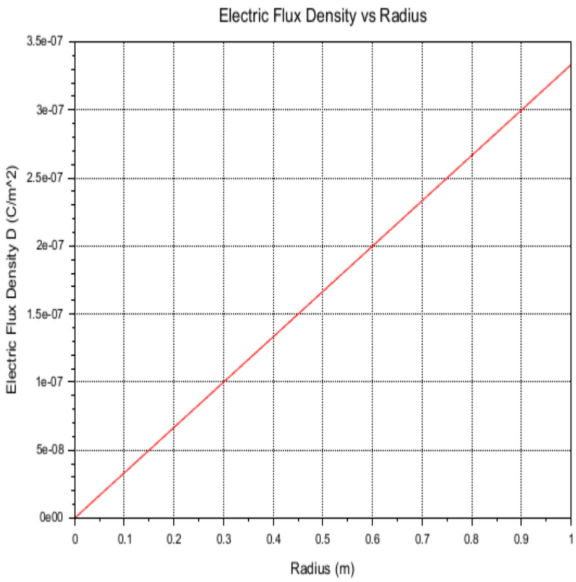
Code:

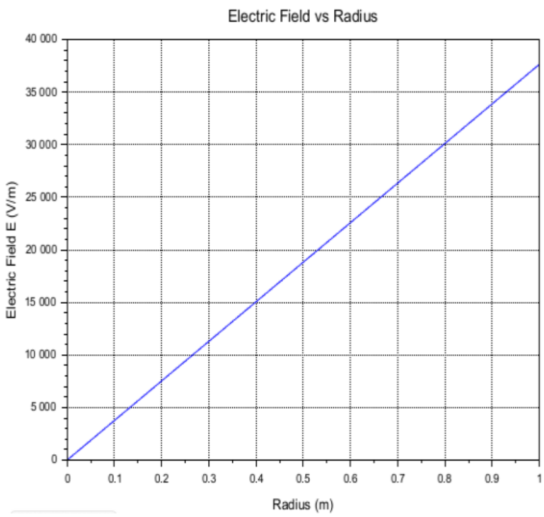
```
clc;
clear;
close;
// Constants
rho0 = 1e-6;       // Charge density (C/m^3)
epsilon = 8.854e-12; // Permittivity of free space (F/m)
R = 1;            // Sphere radius (m)
n = 100;          // Number of points
```

```
// Radius values
r = linspace(0, R, n);

// Initialize arrays
D = zeros(1, n);
E = zeros(1, n);

// Calculate D and E
for i = 1:n
  ri = r(i);

  if ri == 0 then
    D(i) = 0;
    E(i) = 0;
  else
    // Electric Flux Density
    D(i) = (rho0 * ri) / 3;

    // Electric Field
    E(i) = D(i) / epsilon;
  end
end

//-------------------------
// Plot D vs Radius
//-------------------------
scf(1);
plot(r, D, 'r');
xgrid();
xlabel("Radius (m)");
ylabel("Electric Flux Density D (C/m^2)");
title("Electric Flux Density vs Radius");

//-------------------------
// Plot E vs Radius
//-------------------------
scf(2);
plot(r, E, 'b');
xgrid();
xlabel("Radius (m)");
```

```
ylabel("Electric Field E (V/m)");
title("Electric Field vs Radius");
```

Output:





Results and Observations Table

r (m)	$D_{\text{theoretical}}$ (C/m²)	$D_{\text{simulated}}$ (C/m²)	$E_{\text{theoretical}}$ (V/m)	$E_{\text{simulated}}$ (V/m)
0.00	0.00	0.00	0.00	0.00
0.25	8.33×10^{-8}	8.33×10^{-8}	9.41×10^{3}	9.41×10^{3}
0.50	1.67×10^{-7}	1.67×10^{-7}	1.88×10^{4}	1.88×10^{4}
0.75	2.50×10^{-7}	2.50×10^{-7}	2.82×10^{4}	2.82×10^{4}
1.00	3.33×10^{-7}	3.33×10^{-7}	3.76×10^{4}	3.76×10^{4}

Case 2:

For a sphere with a linearly increasing charge density ρv=ρ0r.derive the expressions for the electric field (E) and electric flux density (D). Plot their variations inside the sphere and analyze the trends.

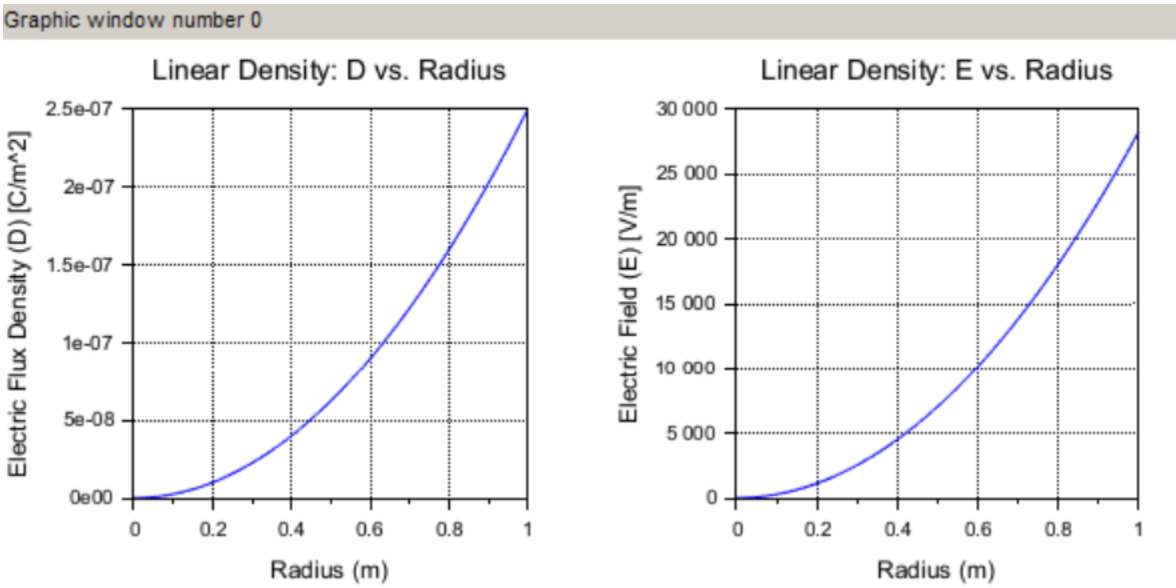
Code:

```
clc;
clear;
clf();
```

```
// Parameters
rho0 = 1e-6;        // Charge density coefficient (C/m^4)
epsilon = 8.854e-12;  // Permittivity of free space (F/m)
R = 1;             // Radius of sphere (m)
n = 100;           // Number of points

// Radius values
r = linspace(0, R, n);

// Initialize arrays
D = zeros(1, n);
E = zeros(1, n);

// Calculate D and E
for i = 1:n

  ri = r(i);

  // Enclosed charge
  Q_enclosed = rho0 * %pi * ri^4;

  if ri == 0 then
    D(i) = 0;
  else
    D(i) = Q_enclosed / (4 * %pi * ri^2);
  end

  E(i) = D(i) / epsilon;

end

// Display values
disp("At Center (r = 0)");
disp(D(1), "Electric Flux Density D = ");
disp(E(1), "Electric Field E = ");

disp("At Surface (r = R)");
disp(D($), "Electric Flux Density D = ");
disp(E($), "Electric Field E = ");
```

```
// Plot Electric Flux Density
subplot(1,2,1);
plot(r, D);
xtitle("Linear Density: D vs. Radius", ...
    "Radius (m)", ...
    "Electric Flux Density (D) [C/m^2]");
xgrid();

// Plot Electric Field
subplot(1,2,2);
plot(r, E);
xtitle("Linear Density: E vs. Radius", ...
    "Radius (m)", ...
    "Electric Field (E) [V/m]");
xgrid();
```

Output:



Results and Observations Table

r (m)	$D_{\text{theoretical}}$ (C/m²)	$D_{\text{simulated}}$ (C/m²)	$E_{\text{theoretical}}$ (V/m)	$E_{\text{simulated}}$ (V/m)
0.00	0.00	0.00	0.00	0.00
0.25	1.56×10^{-8}	1.56×10^{-8}	1.76×10^{3}	1.76×10^{3}
0.50	6.25×10^{-8}	6.25×10^{-8}	7.06×10^{3}	7.06×10^{3}
0.75	1.41×10^{-7}	1.41×10^{-7}	1.59×10^{4}	1.59×10^{4}
1.00	2.50×10^{-7}	2.50×10^{-7}	2.82×10^{4}	2.82×10^{4}

Case 3:

A sphere of radius R=2 m has a uniform charge density ρ=3×10^−6 C/m^3

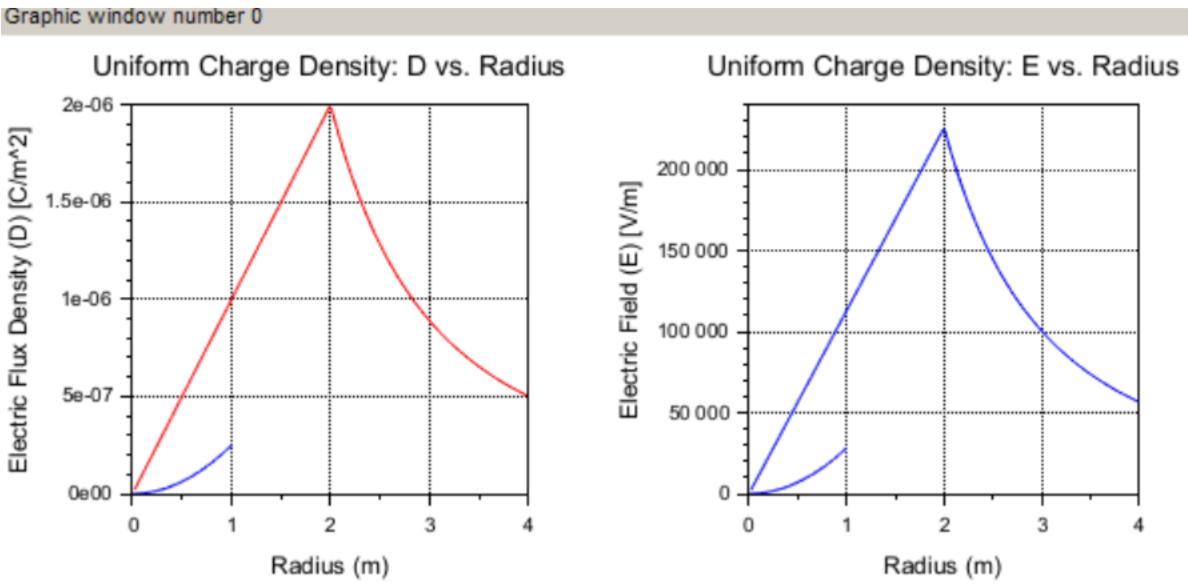
1. **Calculate and plot the electric flux density D(r) and electric field E(r) as a function of radial distance r, ranging from 0 to 4 meters.**
2. **Discuss the behaviour of the electric field inside and outside the sphere.**

Code:

```
rho0 = 3e-6; // Uniform charge density (C/m^3)
epsilon = 8.854e-12; // Permittivity of free space (F/m)
R = 2; // Radius of the sphere (m)
n = 200; // Number of steps for plotting
r = linspace(0, 4, n); // Radial distances from 0 to 4 meters

// Initialize arrays for D and E
D = zeros(1, n);
E = zeros(1, n);

// Calculate the total charge enclosed in the sphere
Q_total = rho0 * (4 * %pi * R^3 / 3);

// Calculate D and E
for i = 1:n
  ri = r(i); // Current radius

  // Enclosed charge for r < R (inside the sphere)
  if ri < R then
    Q_enclosed = rho0 * (4 * %pi * ri^3 / 3);
  // Enclosed charge for r >= R (outside the sphere)
  else
    Q_enclosed = Q_total;
```

```
  end

  // Electric Flux Density D
  D(i) = Q_enclosed / (4 * %pi * ri^2);

  // Electric Field E = D / epsilon
  E(i) = D(i) / epsilon;
end
// Plotting
subplot(1, 2, 1);
plot(r, D, "r");
xlabel("Radius (m)");
ylabel("Electric Flux Density (D) [C/m^2]");
title("Uniform Charge Density: D vs. Radius");
subplot(1, 2, 2);
plot(r, E, "b");
xlabel("Radius (m)");
ylabel("Electric Field (E) [V/m]");
title("Uniform Charge Density: E vs. Radius");
```

Output :



Results and Observations Table

r (m)	$D_{theoretical}$ (C/m²)	$D_{simulated}$ (C/m²)	$E_{theoretical}$ (V/m)	$E_{simulated}$ (V/m)
0.00	0.00	0.00	0.00	0.00
1.00	1.00×10^{-6}	1.00×10^{-6}	1.13×10^{5}	1.13×10^{5}
2.00	2.00×10^{-6}	2.00×10^{-6}	2.26×10^{5}	2.26×10^{5}
3.00	8.89×10^{-7}	8.89×10^{-7}	1.00×10^{5}	1.00×10^{5}
4.00	5.00×10^{-7}	5.00×10^{-7}	5.65×10^{4}	5.65×10^{4}

Result:

Thus, the program was verified for electric field and electric flux density in sphere using SCILAB was successfully.